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Grid World Value Iteration

Overview

This project implements the Value Iteration algorithm for a Grid World Markov Decision Process (MDP). The program computes optimal state values, extracts the optimal policy, and displays the results using a graphical interface.

Requirements

Make sure Python is installed on your system.

Required libraries:

- tkinter (usually comes with Python)
- math (built-in)
- matplotlib (only used for saving images)

If matplotlib is not installed, run:
pip install matplotlib

Files Included

- MDP.py (main program)
- gridworld_easy.txt
- gridworld_medium.txt
- gridworld_hard.txt

How to Run the Program

Step 1: Open terminal or command prompt

Step 2: Navigate to the project folder

Step 3: Run the program:
python MDP.py

```
PROBLEMS 1   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   AZURE

PS C:\Users\banja\OneDrive\Desktop\Project1_CSCE4201\gridworld-value-iteration> python MDP.py
```

Step 4: Enter the gridworld file name when prompted:
Example:
gridworld_hard-1.txt

```
PROBLEMS 1   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   AZURE

PS C:\Users\banja\OneDrive\Desktop\Project1_CSCE4201\gridworld-value-iteration> python MDP.py
Enter the gridworld spec
gridworld_hard-1.txt
```

Program Execution

1. The program first displays the initial grid world.
2. Close the window to continue.
3. The program runs Value Iteration for:

Note: It asks to press enter to continue after each Iteration.

- $\gamma = 0.90$
- $\gamma = 0.95$
- $\gamma = 0.99$

4. For each discount factor:
    - The optimal values and policy are computed
    - The final policy is displayed in a GUI window
    - An image is automatically saved

Output

```
PS C:\Users\banja\OneDrive\Desktop\Project1_CSCE4201\gridworld-value-iteration> python MDP.py
Enter the gridworld spec
gridworld_hard-1.txt

Showing GUI for gamma = 0.9
Gamma = 0.9 converged in 29 iterations
Saved policy image: policy_gamma_0_9.png
Iterations to convergence: 29
Start-state action: UP
Observation: nearest + terminal at distance 10, nearest - terminal at distance 7.
Interpretation: environment places early risk near the start.
Press Enter to continue...

Showing GUI for gamma = 0.95
Gamma = 0.95 converged in 33 iterations
Saved policy image: policy_gamma_0_95.png
Iterations to convergence: 33
Start-state action: UP
Observation: nearest + terminal at distance 10, nearest - terminal at distance 7.
Interpretation: environment places early risk near the start.
Press Enter to continue...

Showing GUI for gamma = 0.99
Gamma = 0.99 converged in 40 iterations
Saved policy image: policy_gamma_0_99.png
Iterations to convergence: 40
Start-state action: UP
Observation: nearest + terminal at distance 10, nearest - terminal at distance 7.
Interpretation: environment places early risk near the start.
Press Enter to continue...
PS C:\Users\banja\OneDrive\Desktop\Project1_CSCE4201\gridworld-value-iteration>
```

The program generates:

- Final policy visualization for each gamma value
- Image files:
    - policy_gamma_0_9.png
    - policy_gamma_0_95.png

- policy_gamma_0_99.png